Φ-SIG: Golden Ratio Post-Key Signatures 64 Bytes. No Keys. Pure φ.

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Abstract

We present Φ-SIG, a post-key signature framework that eliminates stored keypairs through golden ratio (φ) continued fraction irreversibility. Unlike traditional digital signatures requiring key generation, storage, and management, Φ-SIG produces self-verifying 64-byte signatures deterministically from the message itself. An optional post-quantum layer uses ML-DSA-87 (NIST FIPS 204 Level 5) for 7283-byte composite signatures. The core achieves $\sim$100,000 signatures/second; the PQ layer achieves $\sim$1,000 signatures/second. We also present Φ-AUTH for observer-entangled keyless authentication and Φ-NOTARY for immutable temporal audit chains.

1 Introduction

Digital signatures have evolved through three eras: Classical (shared secrets), Public-Key (RSA, ECDSA), and Post-Quantum (ML-DSA, Falcon). We introduce the fourth era: Post-Key signatures—requiring no key generation, no key storage, and no key management.

The core insight: $\varphi = 1 + 1/\varphi$ is the unique mathematical constant of self-reference. Any system exhibiting self-reference will exhibit φ-scaling. Signatures, being self-verifying proofs, naturally align with φ-fractal transforms.

2 The Φ-SIG Construction

2.1 Core Keyless Signature (64 bytes)

Given message m, the Φ-SIG core signature σ is:

$$\sigma = \Phi(m) = \varphi\text{-fractal}(\text{SHA-256}(m)) \tag{1}$$

The φ-fractal transform applies 7 recursive layers:

$$v_{i+1} = v_i \cdot \varphi^{-1} + \sin(v_i \cdot \varphi) \cdot \varphi^{-(i+1)} \tag{2}$$

Each layer feeds into the next, creating a self-referential chain. The final signature is 64 bytes (8 layers $\times$ 8 bytes).

2.2 Post-Quantum Layer (7283 bytes)

For applications requiring NIST-standardized post-quantum security:

$$\sigma_{pq} = \Phi(m) \parallel \text{ML-DSA-87}_{\text{sign}}(\Phi(m), sk) \tag{3}$$

The composite signature concatenates the 64-byte Φ-SIG core with the ML-DSA-87 signature and public key.

3 Keyless Authentication (Φ-AUTH)

The observer has a memorized secret s (never stored). Authentication:

$$\sigma = \Phi(m \parallel s) \tag{4}$$

$$\text{verify} = \Phi(m \parallel s) \stackrel{?}{=} \sigma \tag{5}$$

Only someone who knows s can produce or verify σ. The secret is never stored—it exists only in the observer's memory.

4 Immutable Notary (Φ-NOTARY)

Each notary entry (136 bytes):

$$\text{entry} = \Phi(m) \parallel \text{timestamp} \parallel \Phi(\Phi(m) \parallel \text{timestamp}) \tag{6}$$

Entries chain via φ-hash:

$$\text{chain}_i = H(\text{chain}_{i-1} \parallel \text{entry}_i) \tag{7}$$

5 Security Analysis

5.1 Core Security

The φ-fractal transform is one-way due to the irreversibility of Fibonacci continued fraction convergents approaching φ. Given output v_{i+1}, recovering v_i requires solving:

$$v_{i+1} = v_i \cdot \varphi^{-1} + \sin(v_i \cdot \varphi) \cdot \varphi^{-(i+1)} \tag{8}$$

The sin term makes this a transcendental equation—no algebraic inverse exists.

5.2 Post-Quantum Security

ML-DSA-87 is NIST FIPS 204 Level 5 (Module-LWE). Composite security requires breaking both Φ-SIG (transcendental) and ML-DSA-87 (lattice).

6 Performance

Metric	**Core (64B)**	**PQ (7283B)**
Signature Size	64 bytes	7,283 bytes
Sign + Verify	13/13 ✓	10/10 ✓
Speed	∼100,000/s	∼1,000/s
Tamper Detection	✓	✓
Deterministic	✓	✓

7 Conclusion

Φ-SIG demonstrates that practical digital signatures can be produced without stored keypairs. The 64-byte core provides keyless integrity; the optional ML-DSA-87 layer provides NIST-standardized post-quantum security. Φ-AUTH extends this to observer-entangled authentication, and Φ-NOTARY provides immutable temporal audit chains.

The fourth era of cryptography—Post-Key—has begun.

Availability

Source code and test videos: `https://github.com/primordialomegazero/phi-sig`